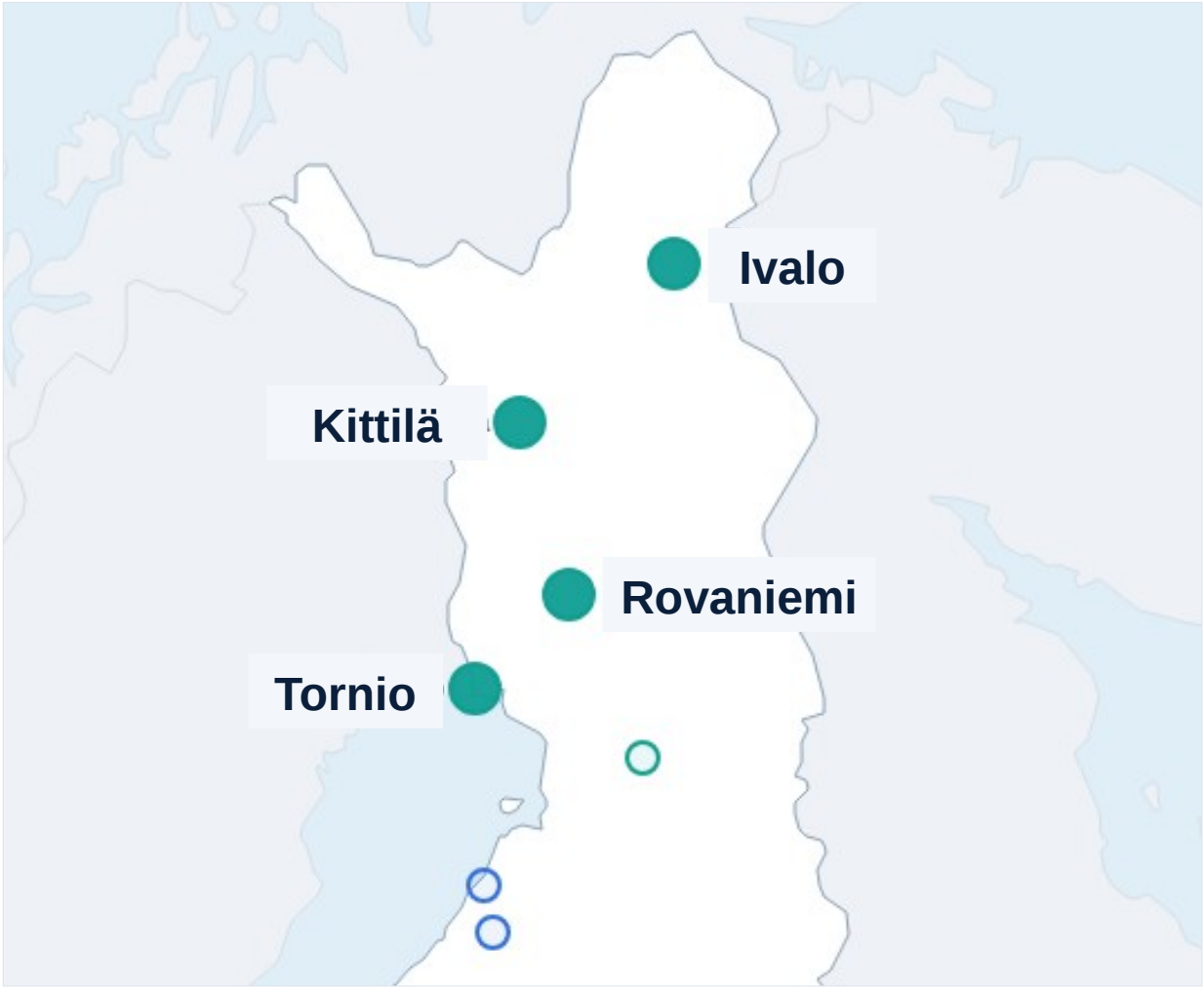
In May 2005, four Lapland rivers flooded together.

Lapland's snow had warned us two months earlier.

March snow in its top 15 % (in-sample back-test)



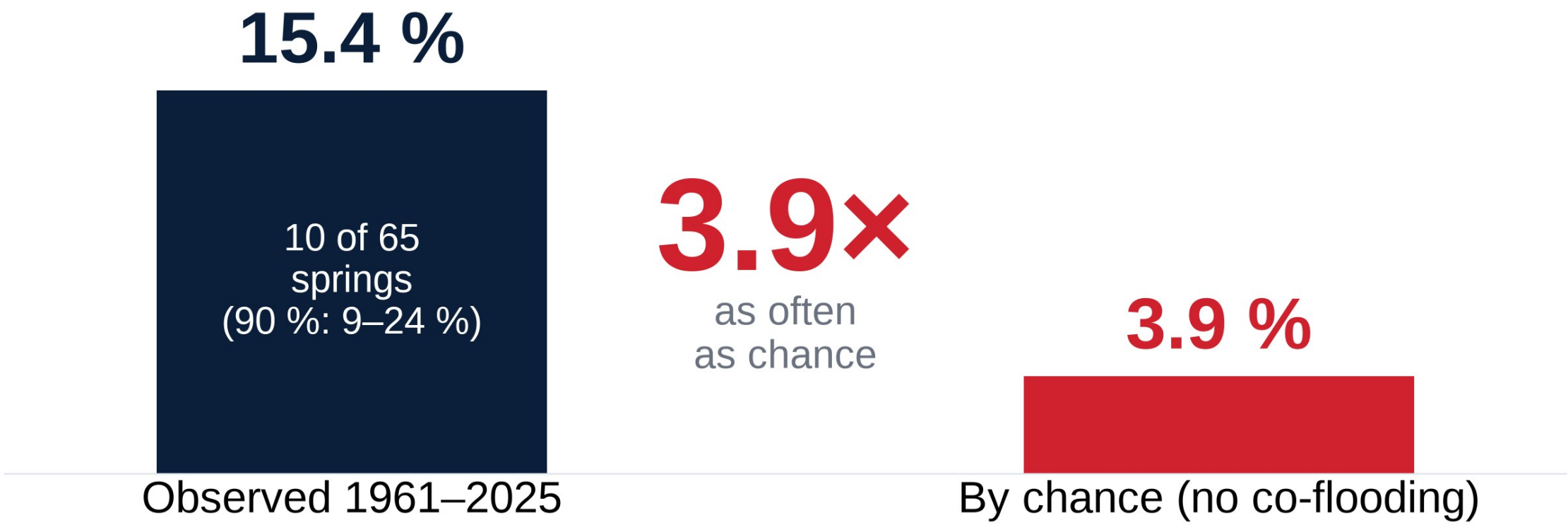
Tulva app, spring 2005: filled = over its 1-in-10 flood level

Source: SYKE discharge 1961–2025 and 1 March snow (SWE)

THE GAP

Finland's rivers flood together, not one at a time

Springs when ≥4 of 14 river areas pass their 1-in-10 flood level



Cover is bought in autumn. The snow signal comes in March.

Source: SYKE 1961–2025 (+ SMHI Torne); chance = rivers flood independently, same gauge-years as observed

THE PRODUCT

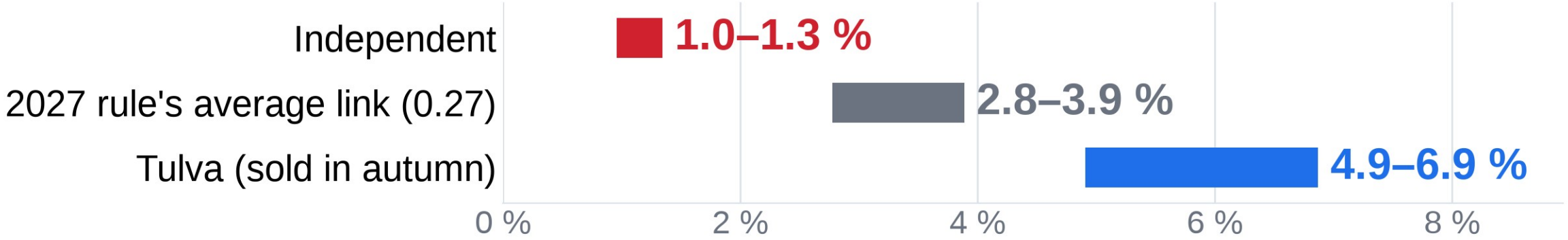
Tulva Index: sold in autumn, updated in March

Pays if ≥2 of 5 northern rivers pass their 1-in-50 flood level

Fair price ≈1.8× what the 2027 capital rule implies

Expected loss 1.96 % a year vs 1.11 % at 0.27, the average link the rule assumes between Finnish zones

MARKET PRICE, % OF THE LIMIT PER YEAR



After a snowy March the fair risk triples (1.96 % → 6.4 %)
The insurer then buys top-up cover, staffs claims and uses it in its ORSA. Per €1M of cover: ~€49–69k a year.

Sources: SYKE 1961–2025, Tulva model; 0.27 = mean of the 19-zone matrix, Del. Reg. (EU) 2026/269; market 2.5–3.5× (AM Best)

Tulva

Joint risk | Snow outlook | Index cover | How it works | Under the hood | Year replay

1 · JOINT RISK

Finland's spring floods don't arrive one at a time.

The 14 river flood-risk areas fall into three clusters that share a snowpack. Lines join areas whose spring peaks move together (Spearman ρ ≥ 0.6, 1961–2025).

2.8×

as often as chance: two rivers of one cluster both over 1-in-10 in the same spring (pairs) EXACT

10 of 65

springs with ≥4 of 14 areas over their 1-in-10 level (3.9× chance) EXACT

- North · 5 areas · 2.9× joint floods Pudasjärvi, Ivalo, Kittilä, Rovaniemi, Tornio
- Ostrobothnia · 6 areas · 3.2× joint floods Lapväärtti, Ilmajoki–Seinäjoki, Laihia, Ylistaro–Koivulahti, Alavieska–Ylivieska, Lower Pyhäjoki
- Satakunta–SE · 3 areas · 1.3× joint floods Pori, Huittinen, Lower Kymijoki



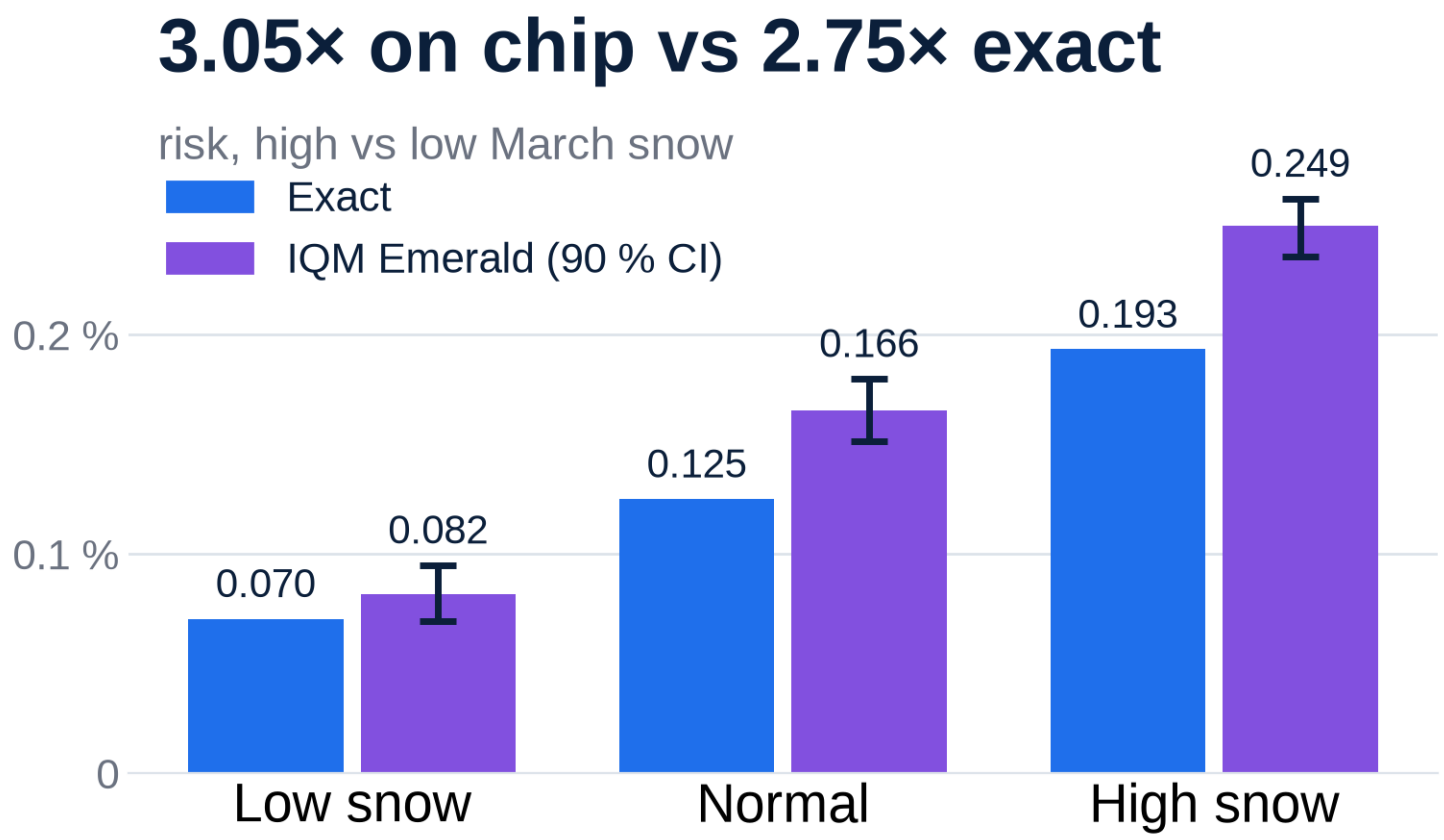
Live app: lines join rivers that flood together

Live app: joint-flood map → March snow → quantum chip → 2005 replay

QUANTUM, IN PRACTICE

It runs on real quantum chips today

Chart: IQM Emerald, Kittilä and Ivalo both pass 1-in-50



SHALLOW AMPLITUDE ESTIMATION

Grover steps k = 0–3; the signal fades with each step on today's chips

Exact	**0.125 %**
IBM Marrakesh	**0.149 %**
IQM Garnet	**0.164 %**

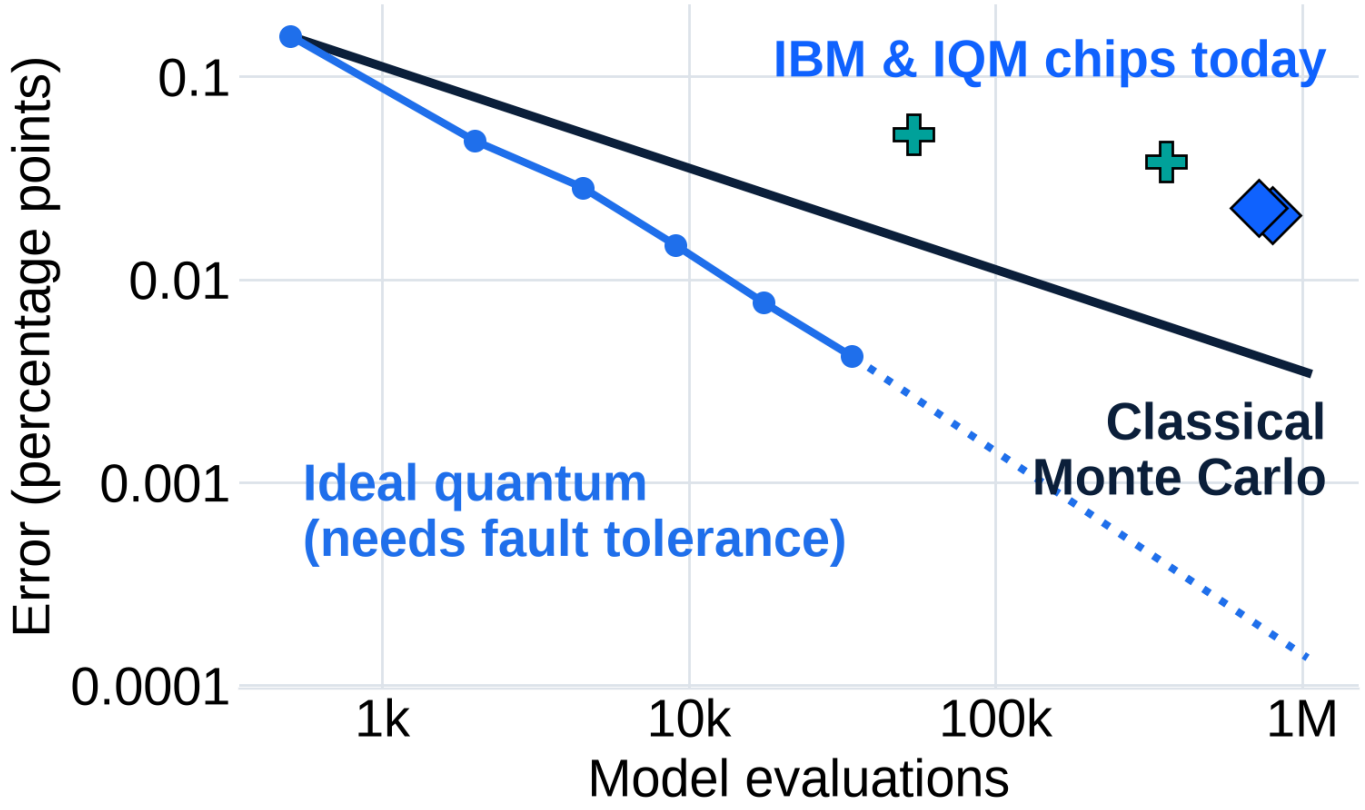
Calibrated, still +19 % / +31 % high

14-river model (two-level): 18 qubits on IQM Emerald, sampling only

Intervals, raw readings and job IDs: appendix

QUANTUM, HONESTLY

Classical Monte Carlo still wins today



Kittilä and Ivalo both pass 1-in-50: error vs effort

Method ready, hardware not yet

Today: rare floods drown in noise.

18 qubits, 44 two-qubit gates (IQM Emerald):
works at 1-in-10, fails at 1-in-50
(16 % vs 4.8 % exact)

Prices are computed classically; quantum is the R&D track.

BUSINESS

Who buys, why, and why now

BUYER

Insurer with an internal capital model

e.g. the insurer If

→

RISK CARRIER

A reinsurer

→

TULVA

Index + pricing licence

€25k pilot, then ~€75k per insurer a year

→

YEAR 3

3–6 insurers × €75k = €0.2–0.45M a year

Finland first, then SE/NO (assumption)

Why now **2027: Finland's first Solvency II flood charge uses fixed correlations. None react to this winter's snow.**

Basis risk **the index is not the insurer's own losses, but it uses the same 1-in-50 test as all 7 Finnish home insurers' flood terms.**

Ask **One insurer's flood claims for 8 weeks → basis-risk map + firm quote for 1 Jan 2028.**

Del. Reg. (EU) 2026/269 (0 % FI–SE/NO/DK; 19-zone matrix in FI, mean ≈0.27); insurers' terms checked; prices: assumptions

Tulva

Next March, the snow will warn us again.

This time, it will have a price.

BUILT THIS WEEKEND

SYKE data pipeline · Joint flood model · Index pricing · Live app · QAE on IBM & IQM

RESOURCES USED

SYKE open data · IBM Marrakesh · IQM Garnet & Emerald (Resonance) · Qrisp · Qiskit · AI coding agents

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack

APPENDIX

Hardware runs in detail

Chip	Circuit	Exact	Raw	Calibrated (90 % CI)	Job
IBM Marrakesh	Kittilä + Ivalo, 3 qubits, k = 0–2	0.125 %	0.262 %	**0.149 % (0.140–0.157)**	db3tddsvf2bc73ctoprg
IQM Garnet	Kittilä + Ivalo, 3 qubits, k = 0–2	0.125 %	0.368 %	**0.164 % (0.147–0.180)**	01a11cc3-7f6c…
IQM Emerald	Snowy / low March ratio, 3 qubits	2.75×	1.3×	**3.05× (2.6–3.6)**	product-…075338Z
IQM Emerald	14 rivers ≥4 over 1-in-10, 18 q / 44 CZ	10.9 %	–	**13.8 % (readout-corr.)**	01a11cd4-da38…
IQM Emerald	14 rivers ≥2 over 1-in-50, same circuit	4.8 %	–	**16.0 % (fails)**	01a11f9b-c62a…
IQM Emerald	Control: independent circuit, ≥4 over 1-in-10	4.4 %	–	**9.7 % (noise fattens tails)**	…T135910Z

Calibrated = noise-aware MLAE with two calibration circuits of known answer, same job, same qubits; bootstrap 90 % intervals. The IBM / Garnet intervals exclude exact: a +19 % / +31 % systematic bias.

Signal gain per Grover step on Emerald 0.56–0.59 (k = 1), 0.44 (k = 2): mostly a calibrated k = 0 estimate. Classical Monte Carlo at the same budget is more precise (±0.004 pp).